

Automated Healthcare Document Processing and Intelligence Platform

Product Requirements Document (PRD)

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1. Product Overview

Healthcare organizations generate and manage large volumes of patient records, laboratory reports, prescriptions, insurance documents, and other medical records every day. Managing these documents manually is time-consuming, prone to errors, and often delays access to critical patient information. Although many healthcare organizations use digital document management systems, efficiently extracting, organizing, and retrieving information from unstructured healthcare documents remains a significant challenge.

The proposed **Healthcare Document Processing and Intelligence Platform** is designed to automate the processing and understanding of healthcare documents. The platform transforms unstructured medical documents into organized, searchable, and easily accessible information through automated information extraction, AI-powered summarization, intelligent search, and document management capabilities. By reducing manual effort and improving access to critical information, it enhances the efficiency of healthcare document management.

To improve transparency and reliability, the platform enables users to verify AI-generated information through references to the original healthcare documents. The system is intended to support healthcare professionals in document management and information retrieval and is not designed to perform medical diagnosis or provide treatment recommendations.

1.1 Problem Statement

The Problem

Healthcare organizations manage large volumes of unstructured healthcare documents, including

Examples of Healthcare Documents

Patient Records	Laboratory Reports
Prescriptions	Discharge Summaries
Insurance Documents	Referral Letters

These documents are typically stored as PDFs, scanned images, or paper records, making critical information difficult to locate, organize, and retrieve efficiently.

Key Challenges

- **Manual Processing:** Reviewing, organizing, and managing healthcare documents requires significant time and effort.
- **Unstructured Information:** Important clinical information is often buried within lengthy and unstructured documents.
- **Slow Information Retrieval:** Finding relevant patient information frequently requires searching through multiple documents.
- **Human Errors:** Manual document handling increases the risk of incorrect, incomplete, or inconsistent information.
- **Poor Accessibility:** Although documents are digitally stored, they often lack intelligent organization and efficient retrieval mechanisms.

Root Cause

Healthcare documents contain valuable clinical information; however, much of this information remains unstructured, making it difficult to organize, retrieve, and utilize efficiently without significant manual effort.

1.2 Proposed Solution

The proposed **Healthcare Document Processing and Intelligence Platform** is an AI-enabled solution that automates the processing and understanding of healthcare documents. The platform transforms unstructured medical documents into structured, searchable, and easily accessible information, enabling healthcare professionals to retrieve relevant information efficiently while reducing manual effort and improving operational efficiency.

Product Flow

Healthcare Document



Upload & Processing



Information Extraction



AI Summarization



Intelligent Search & Question Answering



Verified Results with Source References

How does our solution solve the problem?

Problem	Solution
Manual document reading and processing	OCR + AI-powered information extraction
Critical information buried within documents	Structured information extraction
Slow retrieval of patient information	Intelligent Search + RAG-based retrieval
Lengthy medical reports	AI-generated summarization
Difficulty finding answers across documents	AI-powered Question & Answering
AI reliability and hallucination concerns	Source-backed responses with confidence scores
Extraction inaccuracies	Human verification before finalization

What Value Does It Provide?

Operational Value

- Reduces manual effort involved in processing healthcare documents.
- Accelerates document processing and information retrieval.
- Improves administrative efficiency through workflow automation.

Clinical Value

- Provides faster access to relevant patient information.
- Reduces the time required to review lengthy medical documents.
- Supports healthcare professionals with organized and easily accessible info

Information Value

- Converts unstructured healthcare documents into structured, searchable, and summarized information.
- Enables users to verify AI-generated information using references to the original healthcare documents.

Long-Term Value

- Provides a scalable platform for intelligent healthcare document management.
- Supports future integration with Electronic Health Record (EHR) systems, analytics dashboards, and other healthcare technologies.

2. Project Scope

This section defines the scope of the Healthcare Document Processing and Intelligence Platform by outlining the functionalities included in the project, the features that are explicitly excluded, and the operational boundaries, assumptions, and constraints that guide the system's development.

2.1 In Scope

These are the functionalities that your project is responsible for delivering.

Category	Included Functionality
Document Processing	Upload healthcare documents (PDF and image formats), securely store uploaded documents, and extract text from scanned documents.
Document Intelligence	Automatically classify healthcare documents, extract structured medical information, and generate document summaries.
Search & Information Retrieval	Search processed documents and ask natural-language questions based on uploaded content.
User Interaction	View uploaded documents, extracted information, summaries, and verify AI-generated information against the original documents.
Document Management	Organize, manage, and access processed healthcare documents.

2.2 Out of Scope

Clearly defining exclusions is just as important.

Category	Excluded Functionality
Clinical Decision Support	Medical diagnosis, disease prediction, treatment recommendations, prescription generation, and clinical decision support.
Hospital Management Systems	Appointment scheduling, billing, inventory management, pharmacy management, bed allocation, and complete EMR/EHR management.
Advanced AI Features	Medical image analysis, voice assistants, wearable device integration, real-time patient monitoring, and predictive patient analytics. .
Enterprise Integration	Integration with existing hospital systems, insurance claim processing, government healthcare platforms, and multi-hospital synchronization.

2.3 Project Boundaries

Project boundaries define the scope within which the platform operates by specifying the supported inputs and expected outputs.

Input	Output
PDF healthcare documents	Structured medical information
Scanned healthcare documents	AI-generated document summaries
Medical reports	Searchable document repository
Prescriptions	AI-powered question answering
Insurance documents	Source references for verification
Laboratory reports	Processed healthcare documents with extracted information

2.4 Assumptions

The following assumptions define the expected operating conditions for the platform:

- Healthcare documents are uploaded only by authorized users.
- Uploaded documents are healthcare-related and in supported file formats.
- OCR accuracy depends on the readability and quality of the uploaded documents.
- AI-generated information is intended to support document understanding and information retrieval.
- Users are responsible for verifying AI-generated information before relying on it for operational or clinical purposes.

2.5 Constraints

The following constraints define the technical, ethical, and project limitations of the platform.

Technical Constraints

- OCR performance depends on the quality and readability of uploaded documents.
- AI models may not achieve 100% extraction accuracy.
- Processing time may increase for large or complex healthcare documents.

Ethical Constraints

- The platform is intended to assist healthcare professionals and does not replace human expertise.
- AI-generated information should be verified against the original healthcare documents before use.

Project Constraints

- The project is limited to healthcare document processing and information retrieval.
- The platform does not perform medical diagnosis, treatment recommendations, or clinical decision-making.

3. Users & User Journey

This section identifies the primary and secondary users of the platform, their roles and responsibilities, and how they interact with the system to achieve their objectives. Understanding user needs and workflows helps ensure that the platform is designed to support efficient healthcare document management and information retrieval.

3.1 User Identification

Primary Users

User Role	Responsibilities	Primary Needs
Doctor	Review patient history, laboratory reports, diagnoses, and medications.	Quickly access patient information, view summaries, ask questions, and verify AI-generated information.
Nurse	Review prescriptions, medication schedules, and patient records.	Quickly retrieve patient and medication information.
Medical Records Personnel	Upload, organize, and maintain healthcare documents.	Automated document processing, accurate information extraction, and efficient document search.
Hospital Administrator	Monitor document workflows and operational efficiency.	Reduce administrative workload and improve document accessibility.

Secondary Users

User	Primary Need
Insurance Processing Teams	Access relevant healthcare documents and extracted information to support insurance-related workflows.
Hospital Management	Gain improved visibility into document management processes and operational efficiency.
Medical Researchers (<i>where permitted</i>)	Access structured and searchable healthcare data for research purposes, subject to organizational policies and privacy regulations.

3.2 User Personas

Define what each user is trying to achieve.

Persona 1

Persona	Doctor
Goal	Quickly review a patient's diagnosis, medications, and previous reports before consultation.
Pain Points	Reading lengthy reports, searching multiple documents, limited consultation time.

Persona 2

Persona	Medical Records Personnel
Goal	Process and organize healthcare documents efficiently.
Pain Points	Manual extraction, document organization, searching archived records.

Persona 3

Persona	Nurse
Goal	Quickly verify prescriptions and medication details.
Pain Points	Quickly verify prescriptions and medication details.

3.3 User Journey

Now describe how users interact with the system.

Overall Workflow

```
graph TD; A[None] --> B[Login]; B --> C[Upload Healthcare Document]; C --> D[Document Validation]; D --> E[OCR Processing]; E --> F[Document Classification]; F --> G[Information Extraction]; G --> H[Summary Generation]; H --> I[Document Stored]; I --> J[Search / Ask Questions]; J --> K[View Results]; K --> L[Human Verification]; L --> M[Download / Save];
```

None
Login
↓
Upload Healthcare Document
↓
Document Validation
↓
OCR Processing
↓
Document Classification
↓
Information Extraction
↓
Summary Generation
↓
Document Stored
↓
Search / Ask Questions
↓
View Results
↓
Human Verification
↓
Download / Save

Doctor's Journey

Goal

Retrieve patient information before consultation.

Journey

None
Login
↓
Search Patient
↓
Open Patient Documents
↓
View Summary
↓
Ask Questions
↓
Receive AI Answer with Citations
↓
Review Information
↓
Continue Consultation

Medical Records Officer Journey

Goal

Process newly received documents.

None
Login
↓
Upload PDFs
↓

OCR



Extraction



Review Extracted Information



Approve / Edit



Save



Documents Become Searchable

Nurse Journey

Goal

Access medication information quickly.

None

Login



Search Patient



Open Prescription



Read Summary



View Medication Information



Continue Patient Care

4. Core Features

The platform provides a set of core features to automate the processing and understanding of healthcare documents. These features enable users to upload documents, extract and organize relevant medical information, generate concise summaries, and retrieve information through intelligent search.

4.1 Essential Features

ID	Core Feature	Description	Priority
CF-01	Healthcare Document Ingestion	Allows users to upload supported healthcare documents for automated processing.	Must Have
CF-02	Document Preprocessing & OCR	Converts scanned and image-based documents into machine-readable text and reduces common OCR noise.	Must Have
CF-03	Document Classification	Identifies the type of uploaded healthcare document, such as a laboratory report, prescription, discharge summary, or insurance documents.	Must Have
CF-04	Medical Information Extraction	Extracts relevant medical entities and values from processed healthcare docs.	Must Have
CF-05	AI-Powered Summarization	Generates concise summaries of processed healthcare documents while remaining consistent with the source content.	Must Have
CF-06	Intelligent Search & RAG Assistant	Allows users to search documents and ask natural-language questions using information retrieved from relevant documents.	Must Have
CF-07	Document & Processing Management	Allows users to view uploaded documents, monitor processing status, and access extracted information and summaries.	Must Have
CF-08	Source References & Verification	Displays the exact location within the uploaded document from which AI-generated information was extracted, allowing users to verify responses and improve transparency.	Must Have

4.2 Future Enhancements

The following future enhancements represent additional capabilities that may be incorporated in subsequent versions of the platform. These features are intended to improve usability, expand functionality, and support the evolving needs of healthcare document processing.

ID	Features	Description
FE-01	Smart Migration & Batch Import	Allow healthcare organizations to migrate existing healthcare documents through bulk upload, ZIP import, or folder import with automated processing and progress tracking.
FE-02	AI Confidence & Human Review	Display confidence scores for OCR and AI-extracted information while allowing users to review, edit, and approve extracted data before it becomes searchable.
FE-03	Patient Timeline & Report Comparison	Organize patient documents chronologically and compare multiple reports to highlight changes in laboratory values, medications, or diagnoses over time.
FE-04	Operations & Quality Dashboard	Provide administrators with processing statistics, document quality metrics, OCR confidence levels, failed documents, duplicate detection, and workflow monitoring.

5. Functional Requirements

The following functional requirements define the core functionalities that the system must perform to meet the project objectives. Each requirement specifies the expected behavior of the platform and supports the implementation of the identified core features.

ID	Related Feature	Functional Requirement
FR-001	CF-01	The system shall allow users to upload supported healthcare documents in PDF, JPG, JPEG, and PNG formats.
FR-002	CF-01	The system shall validate uploaded documents before processing.
FR-003	CF-01	The system shall securely store uploaded documents and assign a unique identifier.
FR-004	CF-01	The system shall automatically initiate the document processing workflow after a successful upload.
FR-005	CF-02	The system shall extract machine-readable text from uploaded healthcare documents using OCR techniques.
FR-006	CF-02	The system shall preprocess extracted text to reduce common OCR-related errors and improve text quality.
FR-007	CF-02	The system shall preserve the logical structure of the document, including headings, paragraphs, and tables, where possible.
FR-008	CF-02	The system shall make the processed text available for subsequent document classification and medical information extraction
FR-009	CF-03	The system shall automatically classify uploaded healthcare documents into predefined categories, such as laboratory reports, prescriptions, discharge summaries, and insurance documents.
FR-010	CF-03	The system shall assign the identified document category to the uploaded document for subsequent processing and organization.
FR-011	CF-03	The system shall allow users to view the classified document type after the classification process is completed.

ID	Related Feature	Functional Requirement
FR-012	CF-04	The system shall extract relevant medical information from processed healthcare documents, including patient details, laboratory test names, test values, medications, and diagnoses where applicable.
FR-013	CF-04	The system shall organize the extracted information into a structured format for efficient storage and retrieval.
FR-014	CF-04	The system shall associate the extracted information with its corresponding healthcare document for traceability.
FR-015	CF-04	The system shall make the extracted information available for AI-powered summarization, intelligent search, and other downstream processes.
FR-016	CF-05	The system shall generate concise summaries of processed healthcare documents based on the extracted medical information.
FR-017	CF-05	The system shall present the generated summaries in a clear and easy-to-understand format while preserving the original medical context.
FR-018	CF-05	The system shall make the generated summaries available for users to view alongside the original healthcare document.
FR-019	CF-06	The system shall allow users to search processed healthcare documents using natural-language queries.
FR-020	CF-06	The system shall retrieve relevant information from processed healthcare documents to generate accurate responses to user queries.
FR-021	CF-06	The system shall present AI-generated responses in a clear and easy-to-understand format.
FR-022	CF-06	The system shall provide references to the relevant sections of the original healthcare document that support the generated response.
FR-023	CF-07	The system shall maintain a repository of uploaded healthcare documents for authorized user access.
FR-024	CF-07	The system shall display the processing status of each uploaded document, such as Pending, Processing, Completed, or Failed.

ID	Related Feature	Functional Requirement
FR-025	CF-07	The system shall allow users to view uploaded documents along with their extracted information and generated summaries.
FR-026	CF-07	The system shall allow users to manage uploaded documents, including viewing, deleting, and retrieving previously processed documents.
FR-027	CF-08	The system shall provide references to the original healthcare document for AI-generated summaries, extracted information, and responses.
FR-028	CF-08	The system shall display the corresponding page, section, or highlighted content from the original document to support user verification
FR-029	CF-08	The system shall allow users to access the referenced content directly from the AI-generated output for improved transparency and traceability.

6. Non-Functional Requirements

This section defines the quality attributes and operational standards that the Healthcare Document Processing and Intelligence Platform must satisfy to ensure reliable, secure, efficient, and user-friendly operation. These requirements specify how the system should perform and support the overall quality, usability, and maintainability of the platform.

ID	Category	Requirement
NFR-01	Performance	The system should process uploaded healthcare documents within an acceptable response time under normal operating conditions.
NFR-02	Availability	The platform should be available to users whenever required, except during scheduled maintenance periods.
NFR-03	Security	User authentication and authorization shall be implemented to prevent unauthorized access to healthcare documents.
NFR-04	Data Privacy	Healthcare documents and extracted information shall be stored securely to protect sensitive patient data.
NFR-05	Reliability	The system shall provide consistent processing results and recover gracefully from unexpected failures.
NFR-06	Scalability	The platform should support increasing numbers of users and healthcare documents without significant degradation in performance.
NFR-07	Usability	The user interface shall be intuitive, consistent, and easy to use for healthcare professionals with minimal training.
NFR-08	Maintainability	The system shall be modular and well-structured to support future enhancements and maintenance.
NFR-09	Compatibility	The platform shall support commonly used web browsers and standard healthcare document formats such as PDF and image files.
NFR-10	Transparency	AI-generated information shall include references to the original healthcare documents to enable user verification.

7. Success Criteria

This section defines the measurable outcomes used to evaluate the success of the Healthcare Document Processing and Intelligence Platform. These criteria ensure that the platform meets its intended objectives by improving document processing efficiency, information accessibility, system usability, and overall user satisfaction.

ID	Success Criterion	Measurement
SC-01	Successful document processing	Users can upload and process supported healthcare documents without errors.
SC-02	Accurate information extraction	The platform consistently extracts relevant information from healthcare documents for user review.
SC-03	Effective document summarization	AI-generated summaries accurately reflect the source document and improve readability.
SC-04	Efficient information retrieval	Users can quickly locate relevant information using search and question answering.
SC-05	Transparent AI responses	AI-generated information includes references to the original healthcare documents for verification.
SC-06	Improved user efficiency	The platform reduces the time required to process, search, and review healthcare documents compared to manual workflows.
SC-07	Positive user experience	Users can navigate the platform and complete common tasks with minimal training.
SC-08	Reliable system operation	The platform processes healthcare documents consistently under normal operating conditions.